IP2369_AC Register Description Document

Version/Revision History

Version	Data	Revise contents	Author/reviser
V1.00	2024-3-18	Initial release	IT555
V1.10	2025-7-23	Increase the number of battery strings and other functions	IT555

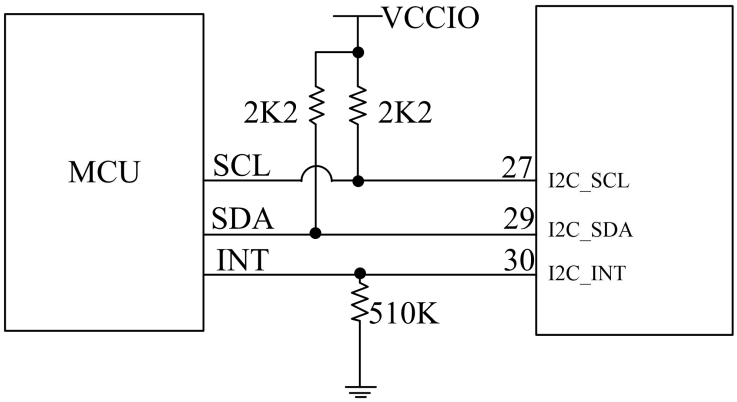


Copyright © 2024, Injoinic Corp.

1 Typical application description

1.1 I2C Connection mode

IP2369 can be used as a slave device, and MCU can read or set the voltage, current, power and other information of IP2369 through the I2C interface. The connection mode of IP2369 I2C is as follows:



1.2 I2C Matters needing attention

- The I2C device address of the IP2369 is written as 0xEA and read as 0xEB. If you need to set other addresses, you can customize them.
- IP2369 I2C communication voltage is 3.3 V, such as the MCU is 5 V voltage, then you need to add level conversion chip, to 3.3 V.
- IP2369 INT Application Instruction: When IP2369 is in sleep mode, it can be awakened by charging and EN. After being awakened, IP2369 actively raises INT. After 100ms, the MCU can perform I2C communication and read/write operations on the registers. Before entering sleep mode, IP2369 will switch to input high impedance to detect the INT status. If it is a high level, it is considered that the MCU does not allow IP2369 to enter sleep mode. If it is a low level, IP2369 enters sleep mode. After the MCU detects that INT is low, it must stop accessing the IC within 16ms;
- IP2369 I2C highest support 250k communication frequency, and consider the clock deviation, suggested that MCU I2C communication clock with 100k to 200k;
- IP2369 a register values for modification, need to the corresponding register values read them first, then the Bit of need to modify and or after the operation, the calculated value written to the register, the other not opened registers cannot be modified. The default value of the register is based on the

Copyright © 2024, Injoinic Corp.

read value. Different IC default values may be different.

- IP2369 I2C communication is real-time data, after receiving the request, the need to interrupt for data preparation, the preparation time is longer, so the MCU I2C communication at the sending address and determine whether receives an ACK after adding 50 us delay (reference I2C application example); It is recommended to read single byte, 100k I2C communication frequency, add 1ms delay between each byte;
- At the end of the I2C read data, after reading the last byte, must give NACK signals, otherwise will think IP2369 continue to read the data, the next clock output will continue to the next data, lead to can't receive the STOP signal, the reading error;
- Reserved registers do not write data at random, do not change the original value, otherwise there will be unexpected results. Register operation must follow the read-Modified-write process. Only the bit to be used can be modified, and other unused bits cannot be modified.
- This document only applies to IP2369_AC models. Other models are invalid.

1.3 I2C 应用示例

After the INT pin of IP2369 remains at a high level for 100ms, the MCU can start I2C communication. First, it can initialize the registers (modify the registers only when necessary for special functions; if no modification is required, the registers can be left unmodified); then it can read the internal information of IP2369 (such as charging and discharging status); finally, it can perform special operations (such as special indicator lights, charging and discharging management, fast charging request management); once the INT pin of the MCU goes low, it needs to stop accessing I2C within 16ms.

For example:

Write the data 0x5A to the 0x05 register

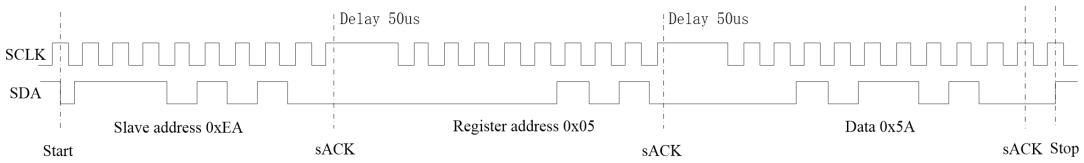


Figure 1 I2C Write 0x05

Read data back from the 0x05 register

Copyright © 2024, Injoinic Corp.

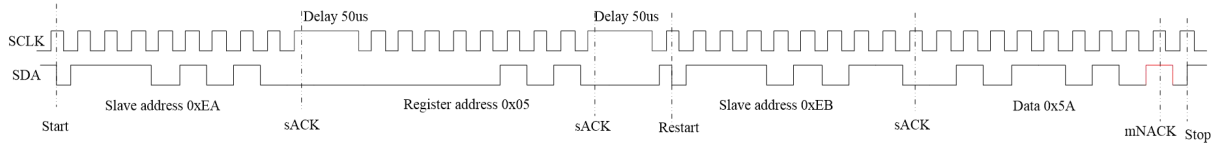


Figure 2 I2C Read 0x05

Copyright © 2024, Injoinic Corp.

2 Register list:

2.1 Read/Write Operation Register

(For some cases, the default values of the registers will change according to the peripheral configuration.)

(0x00) SYS_CTL0（Charge Enable Register）

I2C address 0XEA Register address = 0x00

Bit(s)	Name	Description	R/W	RESET
7	En_LOADOTP	Power-on wake-up to reset register values and enable 0: Do not reset register values 1: Reset register values This bit is not recommended to be set to 0. If modification is needed, the software should regularly reset the default values of the registers, such as after the VINOk VBUOk signals are triggered.	R/W	1
6	En_RESETMCU	MCU reset register Write 1: Reset the register to its default value. After the reset, this bit will automatically revert to 0. After the reset, wait for 2 seconds before performing read/write operations on the register.	R/W	0
5	Reserved			
4	En_Vbus_SinkDPdM	C port input for enabling DM DP fast charging1： Enable 0： disable	R/W	1
3	En _Vbus_SinkPd	C port input enables Pd fast charging1： Enable 0： disable	R/W	1
2	En_Vbus_SinkSCP	C port input enables SCP fast charging (the AC solution requires customizing and opening SCP) 1： Enable 0： disable	R/W	1
1	En_ppath	5V charging and discharging simultaneously function enabled 1： Enable 0： disable (Priority charging)	R/W	1
0	En_Charger	Charging enable (disabled, cannot be charged after this state) 1： Enable 0： disable	R/W	1

Copyright © 2024, Injoinic Corp.

(0x01) SYS_CTL1（VBUS/VOUT1 path MOS enable）

I2C address 0XEA Register address= 0x01

Bit(s)	Name	Description	R/W	RESET
7	En_Vbus_Src_Mos	VBUS port MOS output enable 1：Enable 0：disable	R/W	1
6:5	Reserved			
4	En_Vin_Src_Mos	VOUT1 port MOS output enable 1：Enable 0：disable	R/W	1
3	Reserved			
2	En_Vbus_Sink_Mos	VBUS port MOS input enable 1：Enable 0：disable	R/W	1
1:0	Reserved			

(0x02) SYS_CTL2（Voltage setting of Vset is complete.）

I2C address 0XEA Register address= 0x02

Bit(s)	Name	Description	R/W	RESET
7:0	Vset	Full charge voltage of a single cell Vset = N * 10 mV + 2500 mV (maximum 4.4V)	RW	10101010

(0x03) SYS_CTL3（Iset charging power or current setting）

I2C address 0XEA Register address= 0x03

Bit(s)	Name	Description	R/W	RESET
7:0	Iset	Maximum limit of terminal current of the battery (cannot be set lower than the charging/discharging current) Iset = N * 100mA (maximum value is 9.7A)	R/W	01100001

(0x04) SYS_CTL4（Battery string setting）

I2C address 0XEA Register address= 0x04

Bit(s)	Name	Description	R/W	RESET
7:3	Reserved			
2:0	Bat_Num	Battery string setting Bat_Num = N (maximum 6 strings, minimum 2 strings)	R/W	

Copyright © 2024, Injoinic Corp.

(0x06) SYS_CTL6（Setting of trickle charging current）

I2C address 0XEA Register address= 0x06

Bit(s)	Name	Description	R/W	RESET
7:0	Itk	Setting of trickle charging current Itk=N*50mA	R/W	00000100

(0x08) SYS_CTL8（Stop charging current and recharging threshold setting）

I2C address 0XEA Register address= 0x08

Bit(s)	Name	Description	R/W	RESET
7:4	Istop	Battery charging current setting Istop=N*50mA	R/W	0010
3:2	Vrch	Recharge threshold 00: No recharge function after being fully charged 01: VTRGT － N*0.05 10: VTRGT － N*0.1 11: VTRGT － N*0.2 VTRGT －--- Full charging voltage N---- Number of battery series cells	R/W	10
1	Chg_OvTime	Whether to disable the charging enable after full charge 0: Disable the charging enable automatically after full charge, and retain the charging timeout function 1: Do not disable the charging enable after full charge, and remove the charging timeout function	R/W	0
0	Reserved			

(0x09) SYS_CTL9（Standby enable and low voltage setting）

I2C address 0XEA Register address= 0x09

Bit(s)	Name	Description	R/W	RESET
7	En_Standby	Standby Enable 1: Enable 0: Disable	R/W	1
6	Standby	Write 1 to standby mode, with a single-time effect (bit 7 must be enabled to write 1) 1: Immediately enter standby mode in non-charging condition, 0: Normal process	R/W	0
5	En_BAT_Low	5V Low Voltage Shutdown Enable 0: Disable 1: Enable (After enabling, the shutdown voltage is fixed at 5V. It is only protected by software, and the 涓流 constant current voltage will also change accordingly	R/W	0
4	Reserved			

Copyright © 2024, Injoinic Corp.

3	Standby_Mode	Standby sleep mode selection: (This function is only available in D version IC) 0: Standby sleep mode at 10uA 1: Standby sleep mode at 100uA Note: When the D version chip is loaded, it enters sleep mode only at 100uA.	R/W	ENP is 0 STB is 1
2:0	Reserved			

(0x0A) SYS_CTL10（Battery low voltage setting）

I2C address 0XEA Register address= 0x0A

Bit(s)	Name	Description	R/W	RESET
7:5	Set_BATlow	Battery low voltage setting (the charging current to constant current voltage will change accordingly. Setting below 2.7V only provides software low battery protection) ShutVol = (n+25)*100mV*N n:Set_BATlow N: Number of battery series sections	R/W	101
4	Adjust_En	Battery low voltage recharge setting 0： Subtract the supplementary parameters below Vadjust ShutVol = (n+25)*100mV*N-Vadjust *N 1： Add the supplementary parameters below Vadjust ShutVol = (n+25)*100mV*N+Vadjust *N	R/W	0
3	Reserved			
2:0	Adjust_Data	Vadjust = N*100mV N:Adjust_Data	R/W	0

(0x0B) SYS_CTL11（Output Enable Register）

I2C address 0XEA Register address= 0x0B

Bit(s)	Name	Description	R/W	RESET
7	En_Dc-Dc_Output	Discharge output enable (output is not possible after being disabled) 1: Enable 0: Disable (When there is a charging input on port C, the discharge on port A is not affected by the output enable)	R/W	1
6	En_Vbus_Src_DPdM	C port outputs DP/DM fast charging enable 1: Enable 0: Disable	R/W	1
5	En_Vbus_SrcPd	C port outputs Pd fast charging enable 1: Enable 0: Disable	R/W	1
4	En _Vbus_SrcSCP	C port outputs SCP fast charging enable (AC solution requires customizing and opening SCP) 1: Enable 0: Disable	R/W	1

Copyright © 2024, Injoinic Corp.

3	En_Vin_Src_DPDM	VOUT1 output enables DP/DM fast charging function 1: Enable 0: Disable	R/W	1
2	En_Vin_Src_SCP	Output of Port A: Enable SCP fast charging (The AC solution requires customizing and opening SCP) 1: Enable 0: Disable	R/W	1
1	EN_DRV_EMI	Chip-driven anti-EMI 1: Enable 0: Disable	R/W	0
0	Reserved			

Copyright © 2024, Injoinic Corp.

(0x0C) SYS_CTL12（VBUS Maximum Power Selection Register）

I2C address 0XEA Register address= 0x0C

Bit(s)	Name	Description	R/W	RESET
7：5	Vbus_Src_Power	Vbus output/input power selection: 000：20W 001：27W 010：30W 011：36W 100：45W	R/W	100
4	Reserved			
3	Chg_Power_En	Maximum power input for Vbus is individually configurable and enabled	R/W	0
2:0	Vbus_Snk_Power	Vbus1 input power can be selected independently: 000：20W 001：27W 010：30W 011：36W 100：45W	R/W	101

Copyright © 2024, Injoinic Corp.

(0x0D) SELECT_PDO（Select the charging PDO mode）

You need to read to the 0x35 corresponding position before making a selection. The default selection is the maximum PD gear of the adapter. Once the adapter is unplugged, the configuration becomes invalid. You need to re-identify the adapter's gear position and then make the configuration again.

I2C address 0XEA Register address = 0X0D

Bit(s)	Name	Description	R/W	Reset
7:3	Reserved			
2:0	Pdo_select	Select the charging PDO mode 000: Default setting is the maximum gear position of the adapter. 001：5V 010：9V 011：12V 100：15V 101：20V	R	

(0x10) TEMP_LOOP_CTL（Temperature loop control register）

I2C address 0XEA Register address= 0x10

Bit(s)	Name	Description	R/W	RESET
7	Temp_Loop_En	Temperature loop enable (When enabled, the IC temperature will increase to a certain threshold and then the power will be reduced) 1: Enable 0: Disable	R/W	101
6:3	Reserved			
2:0	Temp_Loop	Temperature loop: High temperature triggers the temperature threshold: Temp_Loop = 80+N*5（℃）	R/W	100

(0x15) CC3CC4_CTL（CC3CC4 output high/low level control register）

I2C address 0XEA Register address= 0x15

Bit(s)	Name	Description	R/W	RESET
7	CC4_OUT	CC4 output (suggested to drive only MOS) 0: Output low level (internal 5.1K pull-down) 1: Output high level (330uA pull-up)	R/W	0
6	CC3_OUT	CC3 output (suggested to drive only MOS) 0: Output low level (internal 5.1K pull-down) 1: Output high level (330uA pull-up)	R/W	0

Copyright © 2024, Injoinic Corp.

5:0	Reserved			

(0x16) STANDBY_TIME（Unstateful sleep time setting register）

I2C address 0XEA Register address= 0x16

Bit(s)	Name	Description	R/W	RESET
7:0	Standby_Time	Standby Time setting Standby_Time = N * 1s (Minimum setting: 2s)	R/W	00001010

(0x17) NTC_CTL（NTC register setting）

I2C address 0XEA Register address= 0x17

Bit(s)	Name	Description	R/W	RESET
7	NTC_EN	NTC enabling control 0: NTC high/low temperature protection is not enabled 1: NTC high/low temperature protection is enabled	R/W	1
6:0	Reserved			

(0x22) TypeC_CTL8（Type-C mode control register）

I2C address 0XEA Register address= 0x22

Bit(s)	Name	Description	R/W	RESET
7:6	Vbus_Mode_Set	Vbus CC mode selection 00：UFP 01：DFP 11：DRP	R/W	11
5:0	Reserved			

Copyright © 2024, Injoinic Corp.

(0x23) TypeC_CTL9（VBUS output Pdo current setting register）

I2C address 0XEA Register address= 0x23

Bit(s)	Name	Description	R/W	RESET
7	En_5VPdo_3A/2.4A	5VPdo Current Setting 1：3A 0：2.4A	R/W	1
6	En_Pps2Pdo_Iset	Pps2 Pdo Current Setting Enable 1: Enable 0: Disable * After enabling, the output power and overcurrent will be based on the set Pdo current. Overcurrent will be 1.1 times the set Pdo current.	R/W	0
5	En_Pps1Pdo_Iset	Pps1 Pdo Current Setting Enable 1: Enable 0: Disable * After enabling, the output power and overcurrent will be based on the set Pdo current. Overcurrent will be 1.1 times the set Pdo current.	R/W	0
4	En_20VPdo_Iset	20VPdo Current Setting Enable 1: Enable 0: Disable * After enabling, the output power and overcurrent will be based on the set Pdo current. Overcurrent will be 1.1 times the set Pdo current.	R/W	0
3	En_15VPdo_Iset	15VPdo Current Setting Enable 1: Enable 0: Disable * After enabling, the output power and overcurrent will be based on the set Pdo current. Overcurrent will be 1.1 times the set Pdo current.	R/W	0
2	En_12VPdo_Iset	12VPdo Current Setting Enable 1: Enable 0: Disable * After enabling, the output power and overcurrent will be based on the set Pdo current. Overcurrent will be 1.1 times the set Pdo current.	R/W	0
1	En_9VPdo_Iset	9VPdo Current Setting Enable 1: Enable 0: Disable * After enabling, the output power and overcurrent will be based on the set Pdo current. Overcurrent will be 1.1 times the set Pdo current.	R/W	0
0	En_5VPdo_Iset	5VPdo Current Setting Enable 1: Enable 0: Disable	R/W	0

Copyright © 2024, Injoinic Corp.

(0x24) TypeC_CTL10（VBUS 5V PDOT current setting register）

I2C address 0XEA Register address= 0x24

Bit(s)	Name	Description	R/W	RESET
7:0	5VPdo_Iset	5VPdo Current Setting 5VPdo=20mA*N	R/W	0x96

(0x25) TypeC_CTL11（VBUS 9V PDOT current setting register）

I2C address 0XEA Register address= 0x25

Bit(s)	Name	Description	R/W	RESET
7:0	9VPdo_Iset	9VPdo Current Setting 9VPdo=20mA*N	R/W	0x96

(0x26) TypeC_CTL12（VBUS 12V PDOT current setting register）

I2C address 0XEA Register address= 0x26

Bit(s)	Name	Description	R/W	RESET
7:0	12VPdo_Iset	12VPdo Current Setting 12VPdo=20mA*N	R/W	0x96

(0x27) TypeC_CTL13（VBUS 15V PDOT current setting register）

I2C address 0XEA Register address= 0x27

Bit(s)	Name	Description	R/W	RESET
7:0	15VPdo_Iset	15VPdo Current Setting 15VPdo=20mA*N	R/W	0x96

(0x28) TypeC_CTL14（VBUS 20VPdo Current Setting Register）

I2C address 0XEA Register address= 0x28

Bit(s)	Name	Description	R/W	RESET
7:0	20VPdo_Iset	20VPdo Current Setting 20VPdo=20mA*N	R/W	0xFA

Copyright © 2024, Injoinic Corp.

(0x29) TypeC_CTL23（VBUS Pps1 Pdo Current Setting Register）

I2C address 0XEA Register address= 0x29

Bit(s)	Name	Description	R/W	RESET
7:0	Pps1Pdo_Iset	Pps1 Pdo Current Setting Pps1 Pdo=50mA*N	R/W	0x3C

(0x2A) TypeC_CTL24（VBUS Pps2 Pdo Current Setting Register）

I2C address 0XEA Register address= 0x2A

Bit(s)	Name	Description	R/W	RESET
7:0	Pps2Pdo_Iset	Pps2 Pdo Current Setting Pps2 Pdo=50mA*N	R/W	0x3C

(0x2B) TypeC_CTL17（VBUS output Pdo configuration register）

I2C address 0XEA Register address= 0x2B

Bit(s)	Name	Description	R/W	RESET
7	Reserved		R/W	R
6	En_Src_Pps2Pdo	Pps2 Pdo Enable 1：Enable 0：disable After disabling, there is no Pps2 Pdo.	R/W	1
5	En_Src_Pps1Pdo	Pps1 Pdo Enable 1：Enable 0：disable After disabling, there is no Pps1 Pdo.	R/W	1
4	En_Src_20VPdo	20VPdo Enable 1：Enable 0：disable After disabling, there is no20V Pdo.	R/W	1
3	En_Src_15VPdo	15VPdo Enable 1：Enable 0：disable After disabling, there is no 15V Pdo.	R/W	1
2	En_Src_12VPdo	12VPdo Enable 1：Enable 0：disable After disabling, there is no 12VPdo.	R/W	1
1	En_Src_9VPdo	9VPdo Enable 1：Enable 0：disable After disabling, there is no 9V Pdo.	R/W	1
0	Reserved		R/W	R

Copyright © 2024, Injoinic Corp.

(0x2C) TYPEC_CTL18（Enabling 10mA current for VBUS PDO requires configuration in conjunction with the current setting register.）

I2C address 0XEA Register address= 0x2C

Bit(s)	Name	Description	R/W	RESET
7:5		Reserved	R/W	0
4	EN_20VPDO_ADD	20VPDO plus 10mA current enable 1：enable 0：disable	R/W	0
3	EN _15VPDO_ADD	15VPDO plus 10mA current enable 1：enable 0：disable	R/W	0
2	EN_12VPDO_ADD	12VPDO plus 10mA current enable 1：enable 0：disable	R/W	0
1	EN_9VPDO_ADD	9VPDO plus 10mA current enable 1：enable 0：disable	R/W	0
0	EN_5VPDO_ADD	5VPDO plus 10mA current enable 1：enable 0：disable	R/W	0

Copyright © 2024, Injoinic Corp.

2.2 Read-only status indication register

Must-read section!!!

When multiple registers represent the same state, each time the lower 8 bits of the register are read, the upper 8 bits and the lower 8 bits of the data will be updated. Therefore, when reading the registers, it is necessary to follow the sequence of reading the lower 8 bits first and then the upper 8 bits to ensure that the data read is the same. For example: When reading the two registers of the BAT terminal voltage, the sequence should be to read 0x50 first, and then read 0x51.

(0x31) STATE_CTL0 (Charging status control register)

I2C address 0XEA Register address = 0X31

Bit(s)	Name	Description	R/W
7:6	Chg_State	Chg_state 00: 5V input for charging 01: High-voltage input for fast charging	R
5	CHG_En	Charging flag 1: Charging state (charging state is considered as long as VbusOk is true) 0: Non-charging state	R
4	CHG_End	Full status flag 1: Charging is complete 0: Charging is not complete	R
3	Output_En	Discharge status flag 1: Discharge status and the output port is already open, with no abnormalities 0: Discharge status but the output port is not open or there is a discharge anomaly	R
2:0	Chg_state	Chg_state 000: Standby 001: Drip Current 010: Constant Current Charging 011: Constant Voltage Charging 100: Charging in Progress (including situations where charging has not been initiated) 101: Fully Charged 110: Charging Timeout	R

Copyright © 2024, Injoinic Corp.

(0x32) STATE_CTL1 (Discharge state control register and MOS state indication register)

I2C address 0XEA Register address = 0X32

Bit(s)	Name	Description	R/W
7	Vbus_Output_State	VBUS output fast charging status 1: Fast charging discharge 0: 5V discharge	R
6	Vin_Output_State	VOUT1 output fast charging status 1: Fast charging discharge 0: 5V discharge	R
5	Mos_Vbus	VBUS output MOS status 1: On state 0: Off state	R
4	Mos_Vin	The output state of VOUT1 port is: 1: On state 0: Off state	R
3	At_Same	Common placement flag 1: In common placement 0: Not in common placement	R
2:0	Reserved		

(0x33) STATE_CTL2 (Input Pd state control register)

I2C address 0XEA Register address = 0X33

Bit(s)	Name	Description	R/W
7	Vbus_Ok	Vbus_Ok 1: Vbus is powered on 0: Vbus is off	R
6	Vbus_Ov	Vbus_Ov 1: Overvoltage of Vbus input 0: No overvoltage of Vbus input	R
5:3	Reserved		
2:0	Chg_Vbus	Charging voltage 110: 20V charging 101: 15V charging 100: 12V charging 011: 9V charging 010: 7V charging 001: 5V charging	R

(0x34) TypeC_STATE（System status indication register）

I2C address 0XEA Register address = 0X34

Copyright © 2024, Injoinic Corp.

Bit(s)	Name	Description	R/W
7	Sink_Ok	Type C Sink input connection flag 1: Valid 0: Invalid	R
6	Src_Ok	Type C Source output connection flag 1: Valid 0: Invalid	R
5	Src_Pd_Ok	Src_Pd_Ok output connection flag 1: Valid 0: Invalid	R
4	Sink_Pd_Ok	Sink_Pd_Ok input connection flag 1: Valid 0: Invalid	R
3	Vbus_Sink_Qc_Ok	The input fast charging validity flags Qc5V and Pd5V do not indicate that the fast charging is successful. 1: Valid 0: Invalid	R
2	Vbus_Src_Qc_Ok	The output fast charging validity flags Qc5V and Pd5V do not indicate that the fast charging is successful. 1: Valid 0: Invalid	R
1:0	Reserved		

(0x35) RECEIVED_PDO（VBUS receiving PDO mode）

I2C address 0XEA Register address = 0X35

Bit(s)	Name	Description	R/W
7:5	Reserved		R
4	PDO_20V	The equipment received PDO20V. 1: Yes 0: No	R
3	PDO_15V	The equipment received PDO15V 1: Yes 0: No	R
2	PDO_12V	The equipment received PDO12V 1: Yes 0: No	R
1	PDO_9V	The equipment received PDO9V 1: Yes 0: No	R
0	PDO_5V	The equipment received PDO5V 1: Yes 0: No	R

Copyright © 2024, Injoinic Corp.

(0x38) STATE_CTL3（System Overcurrent Indicator Register）

I2C address 0XEA Register address = 0X38

Bit(s)	Name	Description	R/W
7:6	Reserved		R
5	Vsys_Oc	The Vsys output overcurrent flag bit needs to be set to 1 to clear it to 0. 1: Vsys output has triggered an overcurrent signal. 0: Vsys output has not triggered an overcurrent signal. If the system continuously detects an overcurrent state for more than two times within 600 milliseconds, it considers the overcurrent to be valid and sets this flag bit to 1. The external master controller can read this flag bit to determine whether an overcurrent anomaly has occurred; from the occurrence of the overcurrent state to the system entering sleep mode, the time is approximately 1.5 seconds.	R
4	Vsys_Scdt	The Vsys output short circuit flag bit needs to be set to 1 to clear it to 0. 1: Vsys output has triggered a short circuit signal 0: Vsys output has not triggered a short circuit signal If the system continuously detects a short circuit state for more than two times within 600 milliseconds, it considers the short circuit to be valid and sets this flag bit to 1. The external master controller can read this flag bit to determine if a short circuit anomaly has occurred; from the occurrence of the short circuit state to the system entering sleep mode, the time is approximately 1.5 seconds.	R
3:0	Reserved		R

(0x39) IC_VERSION（Chip version register）

I2C address 0XEA, register address= 0X39

Bit(s)	Name	Description	R/W
7:0	IC_Version	Chip version 1: B version 2: C version 3: D version ……	R

(0x3A) IC_TEMP（Chip temperature register）

I2C address 0XEA, register address= 0X3A

Bit(s)	Name	Description	R/W
7	Temp_Loop	Temperature loop trigger flag	R
6:0	IC_temp	Chip temperature (° C)	R

Copyright © 2024, Injoinic Corp.

(0x50) BATVADC_DAT0（VBAT Voltage Register）

I2C address 0XEA, register address= 0X50

Bit(s)	Name	Description	R/W
7:0	BATVADC[7:0]	The lower 8 bits of BATVADC data The voltage of VBATPIN	R

(0x51) BATVADC_DAT1（VBAT Voltage Register）

I2C address 0XEA, register address= 0X51

Bit(s)	Name	Description	R/W
7:0	BATVADC[15:8]	The high 8-bit of BATVADC data The voltage of VBATPIN VBAT=BATVADC (mV)	R

(0x52) VsysVADC_DAT0（Vsys Voltage Register）

I2C address 0XEA, register address= 0X52

Bit(s)	Name	Description	R/W
7:0	VsysVADC[7:0]	The lower 8 bits of Vsys voltage data The voltage of VsysPIN	R

(0x53) VsysVADC_DAT1（Vsys Voltage Register）

寄 I2C address 0XEA, register address= 0X53

Bit(s)	Name	Description	R/W
7:0	VsysVADC[15:8]	The high 8 bits of Vsys voltage data The voltage of VsysPIN Vsys= VsysVADC (mV)	R

(0x54) IMOS _C_DAT0（C port MOS current register）

I2C address 0XEA, register address= 0X54

Bit(s)	Name	Description	R/W
7:0	IMos_C[7:0]	The lower 8 bits of the C port MOS current data The input/output current of the C port MOS	R

Copyright © 2024, Injoinic Corp.

(0x55) IMOS_C_DAT1（C port MOS current register）

寄 I2C address 0XEA, register address= 0X55

Bit(s)	Name	Description	R/W
7:0	IMos_C [15:8]	The high 8 bits of the C port MOS current data The C port MOS input/output current The C current = IMos_C (mA) (The MOS internal resistance is calculated as 10mR)	R

(0x56) IMOS _A_DAT0（A Port MOS Current Register）

I2C address 0XEA, register address= 0X56

Bit(s)	Name	Description	R/W
7:0	IMos_A[7:0]	The lower 8 bits of the A port MOS current data The input/output current of the A port MOS	R

(0x57) IMOS _A_DAT1（A Port MOS Current Register）

寄 I2C address 0XEA, register address= 0X57

Bit(s)	Name	Description	R/W
7:0	IMos_A[15:8]	The high 8-bit of the A port MOS current data A port MOS input/output current A port current = IMos_A (mA) (The internal resistance of the MOS is calculated as 10mR)	R

(0x69) TIMENODE1（The first bit of the timestamp register) (The timestamp symbol is an ASCII code character）

I2C address 0XEA, register address= 0x69

Bit(s)	Name	Description	R/W
7:0	TimeNode1	The corresponding value of the first ASCII symbol	R

(0x6A) TIMENODE2（The second bit of the timestamp register）

I2C address 0XEA, register address= 0x6A

Bit(s)	Name	Description	R/W
7:0	TimeNode2	The corresponding value of the second ASCII symbol	R

Copyright © 2024, Injoinic Corp.

(0x6B) TIMENODE3（The third bit of the timestamp register）

I2C address 0XEA, register address= 0x6B

Bit(s)	Name	Description	R/W
7:0	TimeNode3	The corresponding value of the third ASCII symbol	R

(0x6C) TIMENODE4（The fourth bit of the timestamp register）

I2C address 0XEA, register address= 0x6C

Bit(s)	Name	Description	R/W
7:0	TimeNode4	The corresponding value of the fourth ASCII symbol	R

(0x6D) TIMENODE5（The fifth bit of the timestamp register）

I2C address 0XEA, register address= 0x6D

Bit(s)	Name	Description	R/W
7:0	TimeNode5	The corresponding value of the fifth ASCII symbol	R

(0x6E) IBATIADC_DAT0（BAT terminal current register）

I2C address 0XEA, register address= 0x6E

Bit(s)	Name	Description	R/W
7:0	IBATIADC[7:0]	The lower 8 bits of the IBATIADC data for the cell terminal current	R

(0x6F) IBATIADC_DAT1（BAT terminal current register）

I2C address 0XEA, register address= 0x6F

Bit(s)	Name	Description	R/W
7:0	IBATIADC[15:8]	The high 8-bit data of the end current of the battery cell (BATIADC) IBAT= IBATIADC(mA)	R

(0x70) ISYS_IADC_DAT0（Isys terminal current register）

I2C address 0XEA, register address= 0x70

Bit(s)	Name	Description	R/W
7:0	ISYSIADC[7:0]	The lower 8 bits of the VsysIADC data of the IVsys terminal	R

Copyright © 2024, Injoinic Corp.

		current	

(0x71) Isys_IADC_DAT1（Isys terminal current register）

I2C address 0XEA, register address= 0x71

Bit(s)	Name	Description	R/W
7:0	IVsysIADC[15:8]	The high 8 bits of the IVsys terminal current VsysIADC data IVsys = VsysIADC(mA)	R

(0x74) Vsys_POW_DAT0（Vsys power register）

I2C address 0XEA, register address= 0X74

Bit(s)	Name	Description	R/W
7:0	Vsys_POW_ADC [7:0]	The lower 8 bits of the power ADC data at the Vsys	R

(0x75) Vsys_POW _DAT1（Vsys power register）

I2C address 0XEA, register address= 0X75

Bit(s)	Name	Description	R/W
7:0	Vsys_POW_ADC[15:8]	The high 8-bit of the power ADC data at the Vsys Vsys_POW= Vsys_POW_ADC(10mW)	R

(0x77) INTC_IADC _DAT0（NTC output current register）

I2C address 0XEA, register address= 0X77

Bit(s)	Name	Description	R/W
7	NTC_IADC_DAT	0: Output 20uA 1: Output 80uA	R
6:0	Reserved		

(0x78) VGPIO0_NTC_DAT0（VGPIO0_NTC_ADC Voltage Register）

I2C address 0XEA, register address= 0X78

Bit(s)	Name	Description	R/W
7:0	VGPIO0_DAT0 [7:0]	The lower 8 bits of the VGPIO0_ADC data	R

Copyright © 2024, Injoinic Corp.

(0x79) VGPIO0_NTC_DAT1（VGPIO0_NTC_ADC Voltage Register）

I2C address 0XEA, register address= 0X79

Bit(s)	Name	Description	R/W
7:0	VGPIO0_DAT1 [15:8]	The high 8 bits of the VGPIO0_ADC data VGPIO0_DAT = VGPIO0_ADC (in mV) (0 - 3.3V)	R

Copyright © 2024, Injoinic Corp.

4 IMPORTANT NOTICE

INJOINIC TECHNOLOGY and its subsidiaries reserve the right to make corrections, enhancements, improvements and other changes to its semiconductor products and services. Buyers should obtain the latest relevant information before placing orders and should verify that such information is current and complete. All semiconductor products (also referred to herein as “components”) are sold subject to INJOINIC TECHNOLOGY's terms and conditions of sale supplied at the time of order acknowledgment.

INJOINIC TECHNOLOGY assumes no liability for applications assistance or the design of Buyers' products. Buyers are responsible for their products and applications using INJOINIC TECHNOLOGY's components. To minimize the risks associated with Buyers' products and applications, Buyers should provide adequate design and operating safeguards.

Buyer acknowledges and agrees that it is solely responsible for compliance with all legal, regulatory and safety-related requirements concerning its products, and any use of INJOINIC TECHNOLOGY's components in its applications, notwithstanding any applications-related information or support that may be provided by INJOINIC TECHNOLOGY. Buyer represents and agrees that it has all the necessary expertise to create and implement safeguards which anticipate dangerous consequences of failures, monitor failures and their consequences, lessen the likelihood of failures that might cause harm and take appropriate remedial actions. Buyer will fully indemnify INJOINIC TECHNOLOGY and its representatives against any damages arising out of the use of any INJOINIC TECHNOLOGY's components in safety-critical applications.

Reproduction of significant portions of INJOINIC TECHNOLOGY's information in INJOINIC TECHNOLOGY's data books or data sheets is permissible only if reproduction is without alteration and is accompanied by all associated warranties, conditions, limitations, and notices. INJOINIC TECHNOLOGY is not responsible or liable for such altered documentation. Information of third parties may be subject to additional restrictions.

INJOINIC TECHNOLOGY will update this document from time to time. The actual parameters of the product may vary due to different models or other items. This document voids all express and any implied warranties.

Resale of INJOINIC TECHNOLOGY's components or services with statements different from or beyond the parameters stated by INJOINIC TECHNOLOGY for that component or service voids all express and any implied warranties for the associated INJOINIC TECHNOLOGY's component or service and is an unfair and deceptive business practice. INJOINIC TECHNOLOGY is not responsible or liable for any such statements.

Copyright © 2024, Injoinic Corp.